A complete five-coordinate elliptic reference chart

14 November 2026; corrected coefficient table

Abstract

Four quadratic equations in five field coordinates recover a point and its translate, their two ordinate-to-abscissa ratios, and both reciprocal abscissa squares. The proof treats every solution over $\mathbb{Q}(\sqrt{2})$; the apparent vanishing leading coefficient has no solution. Exact rational polynomial certificates verify the identities and exceptional factorization. This edition corrects an omitted fS term in the 13 November 2026 printed coefficient table. The corrected equations are authenticated against all six original rational component rows.

1 The exact equations

Put $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, $D = 11 - 6\rho$ and

$$E : y^2 = x^3 - Dx, \quad P = (a, b), \quad a = \frac{57}{8} - \frac{7}{2}\rho, \quad b = -\frac{165}{8} + \frac{417}{32}\rho.$$

Use coordinates $f, S, B, w, T \in K$ and impose

$$B = f^2, \qquad Q_1 = 0, \qquad Mw + L = 0, \qquad Q_3 = 0. \tag{1}$$

The following expressions specify all coefficients:

$$\begin{aligned} Q_1 &= ST + \left(\frac{26325600}{4905889} - \frac{13327692}{4905889}\rho\right) fT + \left(\frac{279649719}{201141449} - \frac{615250593}{804565796}\rho\right) T \\ &\quad + \left(-\frac{87}{41} + \frac{129}{164}\rho\right) S + \left(\frac{76828374}{4905889} - \frac{48988044}{4905889}\rho\right) f + \frac{21373536}{4905889} - \frac{4638784}{4905889}\rho, \\ Q_3 &= BT + \left(\frac{310419}{658952} - \frac{15327}{329476}\rho\right) fT + \left(-\frac{667923}{658952} - \frac{72481}{329476}\rho\right) fS \\ &\quad + \left(\frac{87}{41} - \frac{129}{164}\rho\right) B + \left(-\frac{11}{49} - \frac{6}{49}\rho\right) T + \frac{570}{2009} + \frac{669}{8036}\rho. \end{aligned}$$

Write $M = B + Af + JS + C$, where

$$\begin{aligned} A &= \frac{23335979380411305}{9142573010269636} - \frac{256685364214285}{4571286505134818}\rho, \\ J &= \frac{44649296511110307}{111996519375803041} - \frac{2886885451827729}{447986077503212164}\rho, \\ C &= \frac{3317607896141091533}{7167777240051394624} + \frac{677690583653850567}{1791944310012848656}\rho. \end{aligned}$$

Finally

$$L = \ell_0 + \ell_f f + \ell_S S + \ell_T T + \ell_B B + \ell_{fS} fS + \ell_{fT} fT,$$

with the seven coefficients listed in Section 5.

Theorem 1.1. *Every K-solution of (1) determines a unique $H \in E(K)$ such that $H, H-P$ are finite with nonzero abscissas, and*

$$f = 1/x(H), \quad S = y(H)/x(H), \quad T = y(H-P)/x(H-P),$$

$$B = 1/x(H)^2, \qquad w = 1/x(H-P)^2.$$

Conversely every such point supplies a solution. At every solution $f \neq 0$ and $M \neq 0$. No rank, scalar, pole or prime-pattern premise is used in this theorem.

2 Recovery without division by a variable leading coefficient

Let $\mathcal{I} = (Q_1(f, S, f^2, T), Q_3(f, S, f^2, T)) \subset K[T, S, f]$. Put

$$g = \frac{((118 + 60\rho)S + 156 + 69\rho/2)T - (156 + 69\rho/2)S + 288 + 128\rho}{252 - 672\rho}.$$

The denominator is a nonzero constant. Write $A_P = a^2 + D$, $B_P = a^2 - D$ and

$$\begin{aligned} L_f &= 2bD(aT - b), & Q_f &= aA_PST + bA_PS - bB_PT + 4a^2D, \\ L_g &= 2bD(aS + b), & Q_g &= -aA_PST - bB_PS + bA_PT - 4a^2D. \end{aligned}$$

The following exact polynomial identities hold modulo $\mathcal{I}$:

$$fS^2 + Df^2 - 1 = 0, \tag{2}$$

$$L_f f - Q_f = 0, \tag{3}$$

$$L_g g^2 - fQ_g = 0, \tag{4}$$

$$D^2 fg^2 + aD(f^2 + g^2) + f\{aST + b(S - T) + D\} = 0, \tag{5}$$

$$M(f, S, f^2)g^4 + f^2 L(f, S, f^2, T) = 0. \tag{6}$$

Each identity is supplied as an explicit pair $U, V \in K[T, S, f]$ whose product sum $UQ_1 + VQ_3$ is its left side. The exact coefficient certificate and its independent checker are specified in Section 6.

Equation (2) gives $f \neq 0$. Set $F = g^2/f$. Equations (3)–(5), dividing only by this already nonzero f, give

$$L_f f = Q_f, \quad L_g F = Q_g, \quad D^2 fF + aD(f + F) + aST + b(S - T) + D = 0.$$

These are exactly the complete fixed-difference chart in [1], including its exceptional leading-coefficient cases. It gives $H = (f^{-1}, S/f)$ and $H - P = (F^{-1}, T/F)$, with $fF \neq 0$. To check the needed nondegeneracy explicitly,

$$C_P = D^2 + 6a^2D - 3a^4 = -\frac{94897379}{4096} + \frac{4195821}{256}\rho \neq 0.$$

The two leading coefficients cannot vanish together, since at $S = -b/a, T = b/a$ one has $Q_f = C_P$. If $L_f = 0$, the chart identities force $F = 1/a$ and the coefficient of f in the remaining equation is $D(a^2 + D)/a \neq 0$. The symmetric case handles $L_g = 0$. Thus no extra inversion of L_f or L_g is assumed.

3 Exclusion of the vanishing coefficient

Suppose $M = 0$. Since $J \neq 0$, we have $S = -(f^2 + Af + C)/J$. Substitution in (2) gives

$$f(f^2 + Af + C)^2 + J^2(Df^2 - 1) = 0.$$

With $f_0 = -291/392 - 17\rho/196$, exact factorization is

$$f(f^2 + Af + C)^2 + J^2(Df^2 - 1) = (f - f_0)^3(f^2 + uf + v),$$

where

$$u = \frac{52622151001413627}{18285146020539272} - \frac{3405676270753831}{9142573010269636}\rho,$$
$$v = -\frac{4974890925663294583793175 4176}{106615613836876012413675285169} + \frac{17501993153580871856729407104}{106615613836876012413675285169}\rho.$$

The discriminant has norm

$$\mathrm{Nm}_{K/\mathbb{Q}}(u^2 - 4v) = \frac{42669856913571796524289}{458737743363289255936}.$$

Its 73-adic valuation is one. The norm of a square in K is a rational square, so the quadratic has no K-root. At the remaining value $f = f_0$, the equation $M = 0$ gives $S = 87/41 - 129\rho/164$. Substitution in Q_1 cancels the entire coefficient of T and leaves

$$-\frac{743972797947}{65974395272} + \frac{139322800357}{16493598818}\rho \neq 0.$$

This is a contradiction, proving $M \neq 0$ on the entire K-solution set. Combining (6) with $Mw + L = 0$ gives $M(f^2w - g^4) = 0$, hence $w = g^4/f^2 = F^2$. This proves the forward direction of Theorem 1.1.

4 Converse and intermediate zero denominators

In the function field of E, the first fixed-difference equation gives

$$T = \frac{bA_PS + 4a^2D + 2b^2Df}{2abDf - aA_PS + bB_P}.$$

The denominator is not the zero function. Substitute this into Q_1, Q_3 with $B = f^2$ and clear its powers. Both numerators are explicit polynomial multiples of $fS^2 + Df^2 - 1$; their quotients are included in the certificate. Thus $Q_1 = Q_3 = 0$ in the function field. On the nonempty open set where $L_g \neq 0$, identities (3)–(5) identify g^2/f with the actual reciprocal abscissa F of $H - P$. Equation (6) gives $MF^2 + L = 0$ there.

All the functions f, S, F, T are regular on the entire open subset in the theorem. Every required equation is polynomial in these regular functions. A function-field identity therefore holds throughout that open subset, including points where a denominator used only in its verification vanishes. This proves the converse and covers all intermediate exceptional cases.

5 Corrected coefficient table

$$
\begin{aligned}
\ell_0 &= -\frac{28738477394907868888067173 71}{43076291407612657992662 4016} - \frac{81557625814705269604141077}{21538145703806328996331 2008}\rho,\\
\ell_f &= \frac{12564555674935844126656 35}{21977699697761560200337 96} + \frac{21448658219217192261719 0}{54944249244403900500844 9}\rho,\\
\ell_S &= -\frac{4766844858863040}{111996519375803041} - \frac{3149175549566448}{111996519375803041}\rho,\\
\ell_T &= \frac{91912184199366551904678 15}{26922682129757911245414001} + \frac{47475418018975282967719917}{21538145703806328996331 2008}\rho,\\
\ell_B &= -\frac{86993}{153664} - \frac{4947}{38416}\rho,\\
\ell_{fS} &= \frac{171357}{235298} + \frac{2519877}{7529536}\rho,\\
\ell_{fT} &= -\frac{47892880950890591270656509}{53845364259515822490828002} - \frac{64726590101578036649524 4709}{17230516563045063197064 96064}\rho.
\end{aligned}
$$

The fS coefficient was omitted in the printed v26 table. Without it, (6) is false, so the omission cannot be ignored as an unproved assertion about a different chart. Restoring it reproduces the original six rational component rows and makes every identity above exact. The corrected 79-variable construction is reconstructed from these equations; this correction is not represented as a comparison with unavailable earlier coefficient-table bytes.

6 Reproducible proof certificates

The directory `computation/quartic/abc79_v27` contains `reference-equations.json`, `reference-certificate.json` and `check_reference.py`. The certificate gives U, V for all five ideal identities, the two forward quotients, and the exact exceptional factorization. Multiplication and collection over $\mathbb{Q}(\sqrt{2})$ verify them without numerical approximation. The standard-library checker imports no symbolic producer. It forms each target independently, checks all seven polynomial identities, reconstructs all six original rational component rows (including the half-coefficient conventions), and verifies the discriminant norm and exceptional substitution. Deliberately omitting the fS term, changing an equation, or corrupting a certificate is rejected. The certificate producer uses exact symbolic arithmetic only to find the identities. This is a computationally certified algebraic proof, not an independent non-author review of itself.

References

[1] *Three quadratic equations for two translated elliptic parameters*, project companion, `fixed_difference_charts_20261110_v18.tex`.

[2] *A 212-variable quartic with additive height comparisons*, project companion, `quartic212_20261113_v25.tex`; the six reference component rows are authenticated against its complete table.